

Who will Trust my Digital Twin? Maybe a Clerk in a Brick and Mortar Fashion Shop

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ABSTRACT

Digital Twin (DT) models mirror the life of physical entities and are adopted to optimize several industrial processes. Although well-established in the industrial fields, one of the most exciting examples of where DTs may be employed is in the MetaVerse, with Human Digital Twins (HDTs). We present a preliminary study that examines the efficacy of HDT-human interactions in the context of a fashion shop. Based on the results obtained involving thirty-two participants in our experiments, we begin a discussion related to the pros and cons of this approach on x-commerce.

Index Terms: Human-centered computing—MetaVerse—Human Digital Twin—X-Commerce

1 INTRODUCTION

The concept of Digital Twins (DTs) appears first in [3], where the author introduces mirror worlds. Three decades later, DTs emerge as components in many types of projects, as they may help reduce the expenditures caused by unexpected disruptions [2]: a DT platform may show how given options would work in specific contexts of use. An interesting new direction of work is the one proposed in [8] where the authors introduce Human Digital Twins (HDTs) as models capable of resembling the inner and outer self of a person and of assisting in everyday tasks. While a fascinating perspective, there is little understanding of how HDTs could be socially accepted. Some works have studied forms of social interactions between Intelligent Virtual Assistants like Alexa and humans in the virtual and augmented realm [4–6], but, to the best of our knowledge, none of them have adopted an HDT perspective. In this work, to focus on HDTs, we carried out a preliminary study in symmetric reality [9]: a virtual entity interacts with a non-virtual one. To this aim, we took as a use-case a brick and mortar fashion shop where a clerk can receive requests from HDTs appearing on a screen. We hence created an avatar 3D-scanning a real person [1] and tested the impact of having this avatar ask a specific question to a human as if the avatar were visiting a shop. Resorting to a group of users, we analyzed such impact, also verifying how such paradigm may be received in an x-commerce setting, as defined in [6]). Our initial results suggest that there might be positive factors in adopting HDTs in brick and

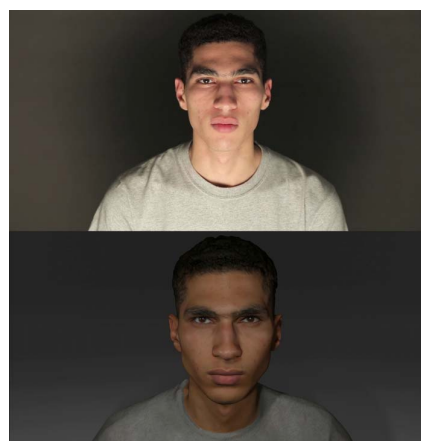


Figure 1: Real customer (top) and corresponding 3D model (bottom).

mortar shops, even if many aspects are yet to explore to make social interactions between humans and HDTs accepted.

2 PRELIMINARY STUDY

We aim at analyzing how social interactions with HDTs are received, considering a brick and mortar shop as a use case.

2.1 Method

We recruited a group of participants to analyze how receiving a communication from an HDT may be accepted. Participants watched three videos: one portraying a real person (customer) and two of the HDT of that person (the two HDTs differ in their voices, one natural and the other faked), each of which lasting 11 seconds. Each participant responded to a survey first providing preliminary impressions of the HDTs and then walking through a process where s/he is asked to identify oneself in a clerk and answer an additional number of questions. Figure 1 shows the real and scanned image of the person who played the role of the customer in our experimental setting.

2.2 Material

After creating the 3D scan of the customer and producing the three videos, the experiment has been administered via the Google Survey platform¹, that also provides the aforementioned videos, as representative of a situation where a clerk interacts with a customer on

¹Link to the survey (Italian)

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a monitor and a trade-off between simplicity and recruiting participants during the COVID19 pandemic. The first video was taken by the independent artist Federica Sasso in a professional setting, fixing the camera in front of the professional model Ahmed Mesbah. His 3D model was then generated using Artec3D Eva scanner [1] and processed/animated with Blender [7], which was also used for the rendering of the two HDT videos.

2.3 Participants

We recruited 32 participants for our experiment, 11 males and 19 females, 21 to 58 years old. The majority fell under the 35 years old threshold ($M = 27.77$, $SD = 8.93$). They were either students, clerks, or professionals with little or no knowledge of digital avatars and 3D models. 15 declared to have no previous experience with digital avatars, while 10 out of 16 occasionally.

2.4 Procedure

After completing consent and demographics questionnaires, participants answered preliminary questions about their occupation (student, worker, fashion shop clerk) and their relationships with digital avatars (specifying if they had at least one contact, the frequency, and their degree of comfort). Afterward, the experiment took place. Each participant was involved in a within-subject session requesting them to identify themselves as fashion clerks and respond to specific customer requests. The first request sounded as follows: “Good morning. I need a shirt to wear with casual gray wool trousers. I usually wear size M, I am available to send you the 3D model of my body if you’d like”. Such request is delivered using three different configurations: (a) natural person using natural voice (NN), artificial 3D model using natural voice (AN), and artificial 3D model using artificial, robotic, distorted voice (AA). After watching each of three videos, a subject is asked to respond to the open question, (“Which product would you propose to the customer?”), in no more than one minute, to facilitate their deep processing state and, at the same time, focusing on the request instead of just on the customer’s visual aspect. Then, a participant responded to different questions concerning the experience, adopting a 7-point Likert scale: (Q1) “Understanding the customer’s request was” (complex vs easy), (Q2) “How natural/spontaneous did you consider the interaction with the customer?” (little vs lot), and (Q3) “During the sale, I felt comfortable”. Finally, each participant answered a few questions concerning the intention to work as a fashion shop assistant in an extended reality setting (measured with a 7-point Likert scale): (I1) “Following this experience, would you work in a virtual store?” and (I2) “If you worked in a virtual store, would you like to interact with HDTs of your customers?”.

2.5 Results and Discussion

The mean and standard deviation of all the Q-questions values are reported in Figure 2. On the one hand, we observe that participants generally preferred watching the video in the NN configuration. On the other, the worst possible configuration is the one that adopts the digital avatar along with an artificial and distorted voice (AA). The AN configuration exhibits lower mean values than the NN one, but this difference does not appear significant. However, we have also to point out that mean overall results for question Q2 never exceed the value of 5. These preliminary results could first suggest that to create a trustworthy and natural interaction between humans and HDTs in a fashion shop, the role of the voice remains fundamental. However, some concerns generally emerge regarding the interaction with an HDT. It is also worth highlighting that answers from items I1 and I2 provide a non-positive attitude on working in an extended reality setting as clerks ($M=3.69$, $SD=1.71$) while a non-negative one in communicating with customers’ HDTs ($M=4.5$, $SD=1.68$). This fact underscores a bias that could have also influenced Q-questions.

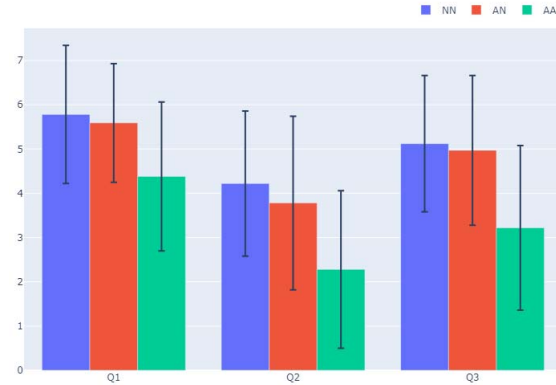


Figure 2: Mean and standard deviations of participants’ answers to Q-questions.

Finally, the majority of participants saw the merit of receiving the 3D model of the customer to better satisfy his request.

3 CONCLUSIONS

We presented a preliminary experiment in which we assessed the possible favor of HDTs as customers of a brick and mortar store. Our results suggest the adoption of such technologies could be positive. Yet, many aspects require further investigation to support any possible social interactions between humans and HDTs. In future works, we aim at improving the experimental design by examining different aspects, contextualizing HDTs and humans in more structured x-commerce settings.

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